

CHAPTER 22

Collecting Sensitive Information with Different Modes of Data Collection

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22.1 INTRODUCTION

The search for better methods to gather information on sensitive topics—questions on private, embarrassing, or criminal matters—has been a long-standing methodological quest. Surveys have always included questions that made some respondents uncomfortable, and since the onset of the AIDS epidemic, the need for data on some very sensitive topics, including sexual behavior and injection drug use, has dramatically increased. Typically such data have been collected in personal surveys, those in which interviewers contact the respondents at their homes or by telephone and ask the questions themselves or provide the respondent with a questionnaire to complete. The survey questions—whether read by the interviewer or by the respondent—are increasingly in electronic form. But, despite the widespread use of these methods to collect sensitive data, the accuracy of the data obtained is still very much in doubt.

With most sensitive topics, these doubts concern underreporting. For example, since its outset, the National Survey of Family Growth (NSFG) has asked respondents in face-to-face interviews about the outcomes of their pregnancies. Responses to these questions are the basis for projections about the number of abortions performed in a given year. Comparisons between these estimates and those derived from data from abortion providers suggests that

the survey respondents report fewer than half of their abortions (Jones and Forrest, 1992). There also is evidence of underreporting in studies of drug use. To cite just one example, Fendrich and Vaughn (1994) show that some respondents who admitted illicit drug use in one round of a longitudinal survey deny or minimize their drug use in subsequent rounds. Discrepancies in the opposite direction are rare.

Underreporting in these cases is likely to reflect both motivational and cognitive factors. Many respondents are reluctant to report these embarrassing or illegal behaviors, but even if they were willing to report them, they would still be susceptible to such additional sources of reporting error as failure of comprehension, lapses of memory, and flawed judgments that could lead to underreporting. Although it might seem unlikely that respondents would simply forget incidents of illicit drug use, say, several considerations suggest that such events may be quite forgettable. These events may happen quickly, evoke little emotional reaction, and trigger few discussions with others afterward. Each of these characteristics of the events is likely to produce poor recall.

Underreporting of sensitive behaviors may be the most common problem, but it is not the only one. Reports about sexual behaviors illustrate a more complex pattern of reporting errors. Within a closed population, men and women should report the same aggregate number of opposite-sex sex partners, since members of both sexes are reporting the same sexual pairings. As Smith (1992) has demonstrated, men consistently report more opposite-sex sex partners than do women, a difference that persists even when differences in population sizes are taken into account. The simplest explanation for this difference is that men and women differ in the direction of their reporting errors, with men overstating their partners and women overlooking theirs. These findings suggest that the validity of self-reported data on sexual behavior is far from perfect.

22.1.1 Effects of the Mode of Data Collection

Over the last 20 years the use of computers in the collection of survey data has revolutionized the survey industry and greatly expanded the roster of methods for collecting data in personal interviews. At least six methods can be used to collect self-reported data on sexual behavior, illicit drug use, or other sensitive behaviors:

- Paper and pencil personal interviews (PAPI).
- Paper and pencil self-administered questionnaires (SAQ).
- Walkman-administered questionnaires (ASAQ).
- Computer assisted personal interviews (CAPI).
- Computer assisted self-administered interviews (CASI).
- Audio computer assisted self-administered interviews (ACASI)

We use each of these terms to refer to methods in which an interviewer contacts the respondents and elicits their cooperation in the study. The interviewers either administer the questions themselves or provide the respondents with a paper or electronic questionnaire and transmit the data back to central office. Our research has examined in-person methods of data collection, and this chapter focuses on these methods. As the present monograph makes clear, there are several additional data collection modes available, including those that employ the telephone (Turner, Chapter 23) or home computer (Saris, Chapter 21). We believe that many of our conclusions will generalize to other methods.

Self-Administration

These six modes of data collection differ in several ways. Perhaps the major difference is whether the questions are self-administered or interviewer-administered. A hypothesis guiding much of the survey literature on reports about sensitive topics is that a major source of error is more or less deliberate misreporting. Respondents doubtless want to cooperate by providing accurate information, but they are not immune to other considerations—such as the desire to avoid embarrassment. Sudman and Bradburn (1974) conclude that personal interviews are more affected by self-presentation concerns than are self-administered questionnaires and that this difference is likely to have an especially large effect on reports about sensitive behaviors. Self-administered questionnaires generally obtain higher levels of reporting of sensitive behaviors than do face-to-face interviews (Bradburn, 1983). The advantages of SAQs have been demonstrated in studies of sexual behavior (Boekeloo et al., 1994), illicit drug use (Aquilino, 1994; Aquilino and LoSciuto, 1990; Schober et al., 1992; and Turner, Lessler, and DeVore, 1992), alcohol consumption (Aquilino and LoSciuto, 1990; Hochstim, 1967), and abortion reporting (London and Williams, 1990; Mosher and Duffer, 1994; Mott, 1985).

Table 22.1 summarizes the results from a series of similar studies examining different methods of collecting data on illicit drug use. The table displays the ratio between the percentage of respondents admitting to a given behavior (e.g., lifetime use of marijuana) under interviewer administration of the questions to the corresponding percentage obtained in self-administered questionnaires. The table indicates, for example, that in the Aquilino (1994) study, respondents who completed an in-person paper and pencil interview (i.e., via PAPI) were only 0.67 times as likely to report they had used cocaine in the past month than were respondents who completed a self-administered paper questionnaire (SAQ). This suggests that at least one-third of the recent cocaine users denied their cocaine use in the in-person interview. Doubtless, some cocaine users also denied their recent use in the SAQ.

Several of these same studies also examined reporting of alcohol consumption (e.g., Aquilino, 1994; Aquilino and LoSciuto, 1990; Schober et al., 1992). Generally, the trend for reports on alcohol consumption is similar to that for

Table 22.1. Mode Effects and Reports of Illicit Drug Use: Proportions of Respondents Reporting Drug Use Relative to SAQ

Study	Modes of Data Collection	Drug	Time Frame		
			Past Month	Past Year	Lifetime
Aquilino (1994)	In-person vs. SAQ	Cocaine	0.67	0.75	0.88
		Marijuana	0.83	0.77	0.98
	Telephone vs. SAQ	Cocaine	1.00	1.00	0.76
		Marijuana	0.67	0.62	0.96
Aquilino and LoSciuto (1990)	Telephone vs. SAQ—Whites	Cocaine	0.83	0.85	1.10
		Marijuana	1.00	0.96	1.00
	Telephone vs. SAQ—Blacks	Cocaine	0.60	0.82	0.83
		Marijuana	0.41	0.72	0.80
Gfroerer and Hughes (1992)	Telephone vs. SAQ	Cocaine	—	0.49	0.70
		Marijuana	—	0.65	0.75
Schober et al. (1992)	In-person vs. SAQ	Cocaine	0.60	0.75	0.89
		Marijuana	0.75	0.83	0.99
Turner, Lessler, and Devore (1992)	In-person vs. SAQ	Cocaine	0.42	0.68	0.94
		Marijuana	0.62	0.77	0.95

Note: SAQ refers to self-administered paper questionnaires; in-person refers to face-to-face (paper and pencil) interviews; telephone refers to paper and pencil telephone interviews.

drug use—less drinking is reported when the questions are administered by an interviewer than when they are self-administered, although the pattern is less striking.

Another characteristic of self-administration is that it reduces the salience of the interviewer in the data collection process. Interviewers may divert the respondent's attention from the task of answering the question, or they may help maintain respondent motivation. Regardless of the overall direction, one might expect the interviewer's effect to be reduced when the questions are self-administered relative to when the interviewer administers them (Tourangeau et al., 1997). However, the effect of the interviewers on the answers may be modest (Sudman et al., 1977).

Computerization

A second dimension that distinguishes the six methods of in-person data collection is the computerization of the data collection process. Computerization can have several effects on the data collected. Most obviously, the programs typically prevent errors of administration, thus reducing the propor-

tion of answers that are missing, outside the permissible range, or logically inconsistent with other answers. Computer assistance may also have more subtle effects on data quality. Laptop computers are still a novelty for many respondents, and their use may impress respondents, making the survey seem more important, more objective, or more scientific than when paper questionnaires are used. At the same time computers may make some potential respondents nervous, increasing their reluctance to participate. Resistance to the computer may be especially high when respondent must interact with the computer directly, as with CASI and ACASI (Couper and Rowe, 1996).

Auditory or Visual Presentation of the Questions

A final difference among the methods is whether the questions are presented aurally, visually, or through both modalities. Methods of data collection, such as CAPI or ACASI, in which the questions are read aloud reduce the overall cognitive burden imposed on the respondent and eliminate the requirement that respondents be able to read. This may be especially important within subgroups where literacy problems are common. On the other hand, ACASI may impose cognitive requirements of its own, such as numeracy and minimal typing skills, especially when open-ended items are used. In addition, when the questions are only read aloud, the respondent has less control over the pace of the interview and may be prone to “recency” effects, favoring options presented at the end of the list of permissible answer categories over those presented at the beginning (Krosnick and Alwin, 1987). Auditory presentation (without concurrent visual display) may also overtax the respondent’s listening ability, and the comprehension of even moderately long or complicated questions may suffer as a result. By itself, auditory presentation does not directly affect privacy, since the questions may be presented via earphones or out of the earshot of others.

Particular combinations of these three underlying variables may produce additional differences among the six modes. For example, self-administered paper questionnaires allow respondents to look ahead and go back to earlier items, reducing the effects of question order (Bishop et al., 1988). In principle, CASI and ACASI applications may also allow respondents to “leaf” through the questionnaire, but as a practical matter respondents are less likely to take advantage of this capability on a computer than with a paper questionnaire. Because CASI and ACASI present the items on the screen one at a time, they may increase the time respondents spend on each question relative to an SAQ. For example, respondents may be prevented from rushing through a block of similarly formatted items displayed individually on the screen rather than presented as a block in a paper questionnaire.

Overall Model of the Effect of Mode

Figure 22.1 depicts an overall model of the effects of the different methods of in-person data collection on responses to sensitive questions. The figure shows the main hypothesized causal links among the features of the different modes,

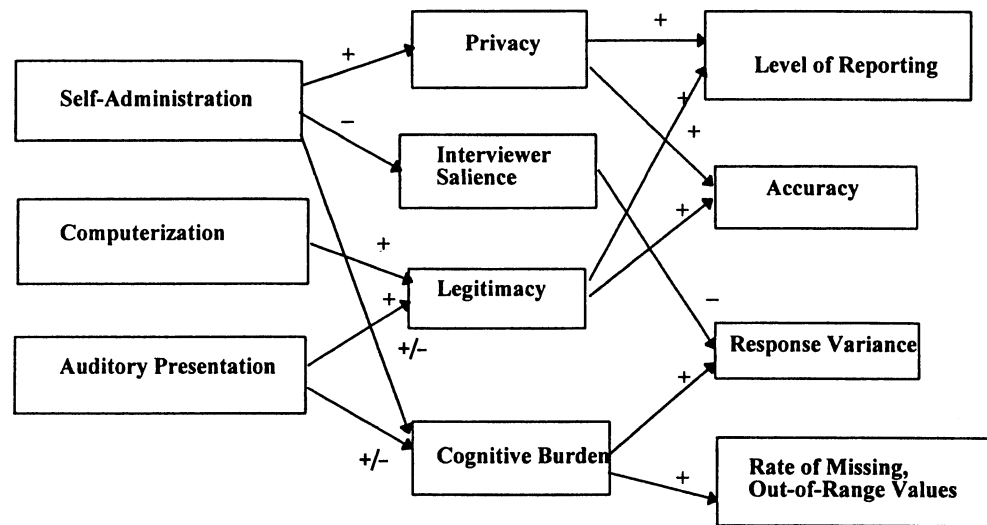


Figure 22.1. Model of mode effects.

the psychological variables that are affected by these features, and the consequences of these psychological variables for the quality of the data that are ultimately collected. According to the model, four variables mediate the effects of data collection mode on data quality—privacy, interviewer saliency, cognitive burden, and sense of legitimacy fostered by the mode. These psychological variables are affected by the method of data collection and in turn affect the properties of the data obtained. The model has not been systematically tested, but it does provide a useful summary of much of what is known about how the method of data collection affects the data that are collected and suggests a number of specific hypotheses for future research. For example, the model incorporates the hypothesis that self-administration increases the perceived privacy of the data collection process and that this effect mediates mode differences in reporting sensitive behaviors. Each link in the diagram corresponds to a similar specific hypothesis about the effects of the mode of data collection on the data that are ultimately collected.

The model is not an attempt to depict all the variables that can affect reporting about sensitive topics, only those that are affected by the method of data collection. Poorly worded questions, difficulties in recall or estimation, and numerous other problems affect reporting without being much affected by the method of data collection. Such sources of mode-invariant error are omitted from the model. The model also omits other aspects of data quality, such as coverage or nonresponse errors. Nicholls, Baker, and Martin (1997) provide a useful summary of the findings on the effect of computerization on data quality.

Figure 22.1 is based on our work with in-person methods of data collection and does not include methods of collecting data by telephone. Some prior

research suggests that telephone data collection is seen by respondents as intermediate in privacy between self-administration and face-to-face interviews (Aquilino, 1994; Hochstim, 1967; Sudman and Bradburn, 1974). However, Smith (1984) reviewed a dozen studies and found no clear indication of any reduction in social desirability effects from telephone data collection (see also Johnson, Hougland, and Clayton, 1989). In addition, telephone interviews clearly rely on auditory presentation of the items. Although the model could be expanded to examine telephone modes, we have chosen to focus on modes of data collection that can be used in the context of in-person data collection.

22.1.2 Types of Items

Our discussion of mode effects has implicitly focused on responses to sensitive questions. Questions about voting, sexual behavior, abortion, sexually transmitted diseases, exercise, smoking, drinking, and illicit drug use have all been considered sensitive. Such questions clearly have socially "correct" answers. In the studies summarized later, we included questions on most of these topics.

The direction of the reporting errors is likely to vary by the topic of the question. Questions about embarrassing or illicit behaviors (e.g., excessive drinking or abortion) are likely to produce underreporting; those about obligatory or widely lauded behaviors (e.g., voting or exercise) are likely to produce overreporting. The findings on sexual behavior are particularly interesting because the direction of the errors seems to vary by the sex of the respondent. The hypothesis that men and women distort their answers in opposite directions (Galbraith et al., 1974; Herold and Way, 1988; Kinsey et al., 1953; Klassen, Williams, and Levitt, 1989; May, Anderson, and Blower, 1989; Ornstein, 1989) is consistent with sex differences in sexual values. Women's attitudes toward sexual behavior are less permissive than men's, and respondents of both sexes show greater disapproval of sexual activity by women than by men (e.g., Klassen et al., 1989).

With nonsensitive behaviors the privacy of the interview will presumably have little or no effect on reporting, so differences among modes related to privacy will be greatly attenuated. Many of the prior reviews of the mode effects literature examine a broader range of items and therefore come to a picture quite different from the one conveyed by Figure 22.1 (e.g., see deLeeuw and van der Zouwen, 1988). In fact one index of a question's degree of sensitivity may be its susceptibility to mode effects, especially to differences between self-administered and interviewer-administered. The data in Table 22.1 show reporting ratios closer to 1.0 (indicating the absence of a self-administration effect) for lifetime drug use than for drug use in the past month. We suspect that recent drug use is more sensitive than drug use in the distant past. Most of the reporting ratios involving lifetime marijuana use are near 1.0. Even political candidates have admitted to trying marijuana.

22.2 STUDIES ON MODE EFFECTS AND REPORTS ON SENSITIVE TOPICS

We have recently conducted a series of studies on sensitive topics, particularly reports about sexual behavior. The first, the Women's Health Study, was carried out as part of a larger effort to test new methods for the National Survey of Family Growth (NSFG). The second examined the effect of several variables, including the mode of data collection, on the disparity between the number of sex partners reported by men and women. The third attempted to determine whether the main source of this disparity was motivated misreporting. A bogus pipeline procedure was used in an attempt to increase motivation to answer truthfully.

22.2.1 Women's Health Study

Since it began in 1971, the NSFG has obtained information on a number of sensitive issues, such as contraceptive practices, pregnancy histories (including fetal and infant deaths), sexually transmitted diseases, and infertility. Our study examined three strategies for improving the accuracy of self-report data in the NSFG.

The first strategy involved increasing the privacy of the data collection. We compared self-administration of the questions with administration by an interviewer. In addition, we investigated the effects of moving the interview to a site outside the respondent's home, away from other family members. Relatively few studies have recorded whether face-to-face interviews on sensitive topics were conducted in private or with other household members able to overhear the respondent's answers. Aquilino (1993) provides an exception. In the studies that have examined this variable, the effects of the privacy of the interview setting have been unclear.

A second strategy we investigated was the use of computer assisted data collection. Early evaluations suggest that computer administration of the items may produce gains similar to those from conventional self-administration (e.g., Waterton and Duffy, 1994) and that computerization itself might have an effect on respondents' willingness to report truthfully (Baker, Bradburn, and Johnson, 1995).

The final strategy we examined was that of placing the interview as a whole, and especially the questions on abortion, in a medical context. We believed that respondents might be more willing to provide truthful answers when the setting underscored the health-related purposes of the interview. (For more detailed descriptions of this study, see Jobe et al., 1997, and Tourangeau et al., 1997.)

Sample

The Women's Health Study comprised a large field experiment, with more than 1000 women being interviewed. (A small comparison sample of 100 men was

included but their data are not presented here.) The sample was selected from an area probability sample of the city of Chicago and from the rosters of two cooperating Chicago-area health clinic. The area probability portion of the sample was a stratified, multistage sample of dwellings in the city of Chicago, selected using standard methods. Screening was used to identify persons in the 15 to 35 age. The clinic component of the sample was drawn from lists of names of women between the ages of 15 and 40 who had had abortions during the preceding year. We selected 2296 persons—1564 from the area sample and 732 from the clinic sample—for the experiment and randomly assigned them to an experimental condition. The study incorporated an after-the-fact permission form procedure in which women were asked to sign a release form giving us access to their medical records at their sources of gynecological care. Forty-eight women from the clinic sample who refused to sign the permission form were dropped from the analysis, and their data were eliminated from the data files.

Study Design

Questionnaires based on the one used in the NSFG were administered to the sample. They included items on abortion, sexual behavior, and illicit drug use and took about an hour to complete. The experiment examined five variables: (1) whether the questionnaire began with a series of questions on medical procedures and conditions or with a series of questions about pregnancy, (2) whether the interview was conducted by a nurse or by a regular NORC interviewer, (3) whether the interview was done at the respondent's home or at a site outside the home, (4) whether the interviewer asked the questions or the questions were self-administered, and (5) whether the data were collected via computer or on paper. The latter two variables were crossed yielding four methods of data collection—CAPI, PAPI, CASI, and SAQ. In an attempt to keep the response rates comparable across conditions, we offered respondents \$40 to complete the interviews outside their homes. This incentive proved over-generous; the outside-the-home group ended up with a higher response rate than the in-home group. The analysis examined a number of outcome variables, including unit and item nonresponse rates under the various experimental conditions, level and accuracy of abortion reporting, and level of reporting on other sensitive topics (e.g., number of sex partners). The main findings involved the effects of self-administration and computerization on reports about sexual behavior, on item nonresponse, and on interviewer variability.

Number of Reported Sex Partners

The questionnaire asked respondents to report the number of their opposite-sex sex partners during the past year, the past five years, and over their lifetimes. The sex partner data are counts, and the distribution of the responses for all three time periods is highly skewed. To compensate for this departure from normality, we added 0.5 to the values and then carried out a logarithmic

transformation prior to performing statistical analyses. For ease of interpretation, the means reported are generally based on untransformed data. The analyses examined the number of sex partners reported by the women respondents as a function of the five experimental factors, controlling for differences across the experimental groups introduced by nonresponse. The standard errors and significance tests took the clustering of the sample into account.

For all three time periods, women who completed self-administered questionnaires reported more sex partners than women who responded to questions administered by an interviewer (see Table 22.2). The effects of self-administration were significant for all three time periods. For the past year, the women who answered self-administered questions reported a mean of 1.72 sex partners versus 1.44 for those who answered interviewer-administered questions. For the five-year period, the corresponding means were 3.87 sex partners versus 2.82. For the lifetime item, the means were 6.51 versus 5.43. No other main effects were significant.

Table 22.2 also displays the proportion of respondents who reported ever having used illicit drugs. The absence of mode differences on drug reporting may reflect the lifetime reference period for the illicit drug use item. As we noted earlier, recent drug use seems more sensitive than drug use in the distant past.

Computerization interacted with the site of the interview to affect the number of sex partners reported. During home interviews, more sex partners were reported by women interviewed using computer assisted questionnaires than by those responding to paper and pencil questionnaires. For women

Table 22.2. Mode Effects in the Women's Health Study

Experimental Group	Mean Reported Sex Partners			Percent Admitting Illicit Use
	Past Year	Past 5 Years	Lifetime	
Self-administered questions	1.72	3.88	6.54	40.9
Conventional (SAQ)	1.56	3.37	6.88	42.5
Computer assisted (CASI)	1.89	4.40	6.25	39.3
Interviewer-administered questions	1.44	2.82	5.43	40.7
Conventional (PAPI)	1.56	2.86	4.58	39.3
Computer assisted (CAPI)	1.36	2.79	6.27	42.2

Source: Tourangeau et al. (1997).

Note: Each mean or proportion is based on approximately 240 interviews; rows for self- and interviewer-administration are based on nearly 500 interviews.

interviewed outside the home, more sex partners were reported on the pencil and paper questionnaires. Bringing computers into the respondents' homes may have fostered a sense of the importance or objectivity of the survey, promoting more complete reporting of sex partners. Outside the home, especially in public places, the computer may make respondents feel conspicuous, inhibiting reporting. Any conclusion must remain tentative at best, since the site variable is confounded with the incentive offered to the respondents.

Item Nonresponse

We analyzed responses to 43 of the questionnaire items administered to the women respondents. These items covered a range of topics—demographics, sexually transmitted diseases, pregnancy history, medical procedures for inducing abortion, illicit drug use, and sexual behavior. Because of the skip patterns in the questionnaire, few respondents were actually administered all 43 of the items: on the average, the respondents were supposed to have answered about 30 of them. Across the experimental treatments, the women answered almost all of the items they were supposed to—an average of 29.4 items, or about 97.4 percent of the total.

Both computerization and self-administration appeared to affect the completeness of the data, as shown in Table 22.3. Computerization increased the proportion of the questions that were answered (98.5% vs. 96.2% for paper questionnaires), and self-administration decreased the proportion (96.2% vs. 98.5% for interviewer administration). In an analysis that fit a simple additive model including the five experimental factors, both of these main effects were significant. A fully saturated five-way model that incorporated all the interactions among the experimental variables revealed an additional interaction effect involving the type of interviewer—the difference between self- and interviewer administration was more striking for the regular field interviewers than for the nurses.

Table 22.3. Proportion of Questions Answered by Method of Data Collection

Group	Mean Proportion	(n)
PAPI	97.8	(261)
SAQ	94.6	(256)
CAPI	99.2	(244)
CASI	97.9	(244)
Self-administered	98.5	(505)
Interviewer-administered	96.2	(500)
Computer assisted	98.6	(488)
Paper and pencil	96.2	(517)

Source: Tourangeau et al. (1997).

Interviewer Effects

One of the potential benefits of self-administration is a reduction of the effect of the interviewers on the answers. When the interviewers merely hand the respondent a questionnaire to be completed (or set up a laptop computer to administer the items), they seem likely to have less effect than when they read the questions and record the answers.

Computerization may also reduce the interviewers' effect on the answers by enforcing a certain level of standardization on the interviewers. We explored these hypotheses by conducting one-way analyses of variance across interviewers for 19 sensitive items, including questions on sex partners, sexually transmitted diseases, specific sexual practices (e.g., oral sex), abortion, and illicit drug use. Separate analyses were carried out for respondents interviewed under each of the four modes of data collection. Because the interviewer assignments were geographically clustered, the ANOVA estimates represent both interviewer effects and the effects of the overall area in which an interviewer's cases were concentrated. The findings are thus suggestive at best, and we make no attempt to test their statistical significance.

Despite these limitations a clear pattern emerged in the data. Using the estimated variance components from the ANOVAs, we estimated values of ρ , the intraclass correlation, for each variable and mode of data collection. (When the estimate of the between-interviewer variance component was negative, we set $\hat{\rho}$ to zero.) Table 22.4 displays the median value of the ρ estimates across the 19 variables, as well as the results for six specific items. Both computerization and self-administration appear to reduce variation across interviewers. For most of the variables, there is no evidence for interviewer effects when the questions were self-administered; the estimate for ρ was typically zero. Similar results were reported in a comparison of CATI and paper and pencil telephone interviews by Mathiowetz and Groves (1984), which found lower estimates of ρ for CATI than for paper and pencil interviews on 18 of 24 variables. Although consistent, these differences were not significant (see also Groves and Magilavy, 1980).

Table 22.4. Estimates of ρ by Method of Data Collection and Variable

	PAPI	SAQ	CAPI	CASI
Median across 19 variables	0.072	0.000	0.016	0.000
One-year sex partners	0.158	0.036	0.000	0.000
Five-year sex partners	0.072	0.000	0.000	0.000
Lifetime sex partners	0.178	0.000	0.021	0.000
Used illicit drugs	0.108	0.003	0.002	0.000
Used marijuana	0.100	0.000	0.013	0.000
Used cocaine	0.100	0.044	0.035	0.000

Source: Tourangeau et al. (1997).

22.2.2 ACASI Experiment

We completed the second experiment on the effects of mode of data collection in the fall of 1994 (Tourangeau and Smith, 1996). This study compared three methods of computer assisted data collection for face-to-face interviewing—CAPI, CASI, and ACASI. The three modes were compared in a cross-sectional sample of respondents from Cook County, IL. The key outcome variables were nonresponse rates and levels of reporting, particularly reporting of illicit drug use and sexual behavior. Our hypotheses were that relative to CAPI data collection, both CASI and ACASI would increase the proportion of respondents admitting illicit drug use and decrease the disparity between the average number of sex partners reported by men and women. More generally, we predicted that by offering greater privacy to the respondents, the two self-administered modes would reduce the effects of social desirability on the answers. Because a large number of partners is seen as undesirable for women and a small number of partners is seen as undesirable for men, we expected self-administration to have opposite effects on the number of sex partners reported by men and women. We did not anticipate a difference in response rates by mode.

The design included two other variables thought to affect the number of sex partners reported. These were the format and context of the sex partner questions. Work described in Bradburn, Sudman, and Associates (1979) had shown that closed-ended items tend to elicit lower levels of reporting of alcohol consumption than open items, at least among those who report drinking at all. To test the effect of the answer options used in the closed questions, we developed two versions of each closed item—one with categories mainly at the low end of the range and a second version with categories concentrated at the high end. As Schwarz and Hippler (1987) have shown, the response categories offered can effect the overall level of the answers (see also Schwarz et al., 1985). Thus we prepared two closed versions and also included an open version of each of the sex partner questions.

Sample

The sample was a multistage area probability sample of dwellings in Cook County, IL. Seventeen of the 32 sample segments (individual blocks or adjoining blocks) were located in the city of Chicago, and the remainder were drawn from the balance of the county. Within those segments we selected more than 1100 housing units (HUs) for a short screening interview to determine whether anyone living there was in the 18–45 age range. Screeners were completed at 975 of the 1069 occupied HUs, for a completion rate of 91.2 percent.

Of the 643 potential respondents selected in the screener, 365 completed the main interview for a final completion rate of 56.8 percent. Unfortunately, the study used a new version of the software, and because of a design flaw in the program, some 79 of these initial interviews were lost. Subsequently the error

was corrected, and 53 of the original cases were reinterviewed under the same experimental conditions to which they had been assigned originally. The results presented here are, except where noted, based on 339 completed cases (including the 53 reinterviewed cases). The results are not appreciably altered if the reinterviews are excluded from the analysis. Although the response rates may seem low, they are comparable to those achieved within large metropolitan areas by major national surveys (e.g., the NSFG).

Experimental Design

Prior to screening, each dwelling was randomly assigned to an experimental condition; the data collection mode variable was crossed with the context and format of the sex partner items, producing a total of 12 conditions. The random assignment was done within segments to ensure that the experimental conditions were not confounded with geographic areas. The main experimental variable was the mode of data collection: Respondents were assigned to data collection by computer assisted personal interview (CAPI), computer assisted self-administered interview (CASI), or audio computer assisted self-administered interview (ACASI). In all cases an interviewer was present throughout the interview. In the CASI and ACASI conditions, interviewers were instructed not to look at the screen but to busy themselves with administrative chores while the respondents completed the questionnaire.

We also examined the effects of prior items on the reporting of sex partners. Just before the sex partner questions, respondents answered one of two sets of questions about their sexual attitudes. One set of items was designed to encourage respondents to report large numbers of sex partners; these questions consisted of statements expressing "permissive" views about sexual activity. The other set was designed to discourage reporting large numbers of partners; these statements in this set expressed more "restrictive" views about sex. We thought that because reports about sexual behavior may be affected by attitudes toward sex, the permissive items would encourage female respondents to admit how many partners they had and that the restrictive items would encourage male respondents to admit how few they had. The open version of the sex partner question simply asked for the number of partners. For example, the one-year question administered to women read: "During the last 12 months, that is, since August/September, 1993, how many men (if any) have you had intercourse with? Please count every partner, even those you only had sex with once." A number was entered as the answer. The closed versions of this question presented the same question followed by a set of response options. The response options presented to one group of respondents were 0, 1, 2, 3, 4, and 5 or more; the options presented to the other group were 0, 1 through 4, 5 through 9, 10 through 49, 50 through 99, and 100 or more. A follow-up item asked respondents in the closed format groups the exact number of partners (when that could not be inferred from the response category they had selected). Respondents received the same version of the sex partner item for all three recall periods and the same set of response options.

Aside from the items on sex partners, the questionnaire included items on a range of sensitive topics, including sexually transmitted diseases, pregnancy, condom use, and illicit drug use. The questions were mostly drawn from existing sources, including the NSFG, the sex supplements to the General Social Surveys, and the National Household Survey of Drug Abuse.

Response Rates

The mode of data collection did not appear to affect the response rates; 55.5 percent of those assigned to ACASI, 57.7 of those assigned to CAPI, and 57.1 percent assigned to CASI data collection completed the interview. We examined whether different sorts of respondents took part under the three modes, comparing the composition of the three groups on sex, age, marital status, and educational attainment. The three groups differed only on educational attainment, with the ACASI respondents reporting a somewhat higher level of education than those in the other mode groups. The differences in reporting by mode described below remain significant even when these differences in educational attainment are taken into account.

Reporting of Sex Partners

Each respondent was asked to report the number of sex partners over three time periods: past year, past five years, and lifetime. We examined the final exact answers to these questions here. Few discrepancies were found between the final and initial answers to the closed questions. Again we transformed the raw counts (by adding 0.5 and taking the log) to reduce the skewness in the data. We also dropped five outliers—two respondents who reported 50 partners in the prior year, one who reported more than 100 partners during the previous five years, and two who reported more than 100 partners in total. Their omission had little effect on the results. We analyzed the transformed reports for each time period by sex, data collection mode, question format, and item context.

For all three time periods, there were significant main effects for the sex of the respondent. As has been found in prior studies, the male respondents reported significantly more partners than their female counterparts (2.9 vs. 1.6 for the past year; 5.2 vs. 2.7 for the past five years; and 8.4 vs. 4.8 over the lifetime). In addition there were significant main effects for all three of the experimental variables (see Table 22.5). For all three time periods, ACASI elicited the highest mean number of reported sex partners. The effect of data collection mode was significant for the lifetime reports and marginally significant for the five-year reports. An additional contrast revealed that for both lifetime and five-year reports, ACASI yielded significantly higher numbers of reported partners, on average, than CASI did. For the one-year and five-year periods, the closed item with the high-response categories elicited the highest level of reporting and the closed item with the low-response categories the lowest. In both cases, the open item produced intermediate levels of reporting. The effect of item format was also significant for both the one-year and

Table 22.5. Mean Reported Sex Partners and *F* Values by Experimental Variables

	One-Year Partners			Five-Year Partners			Lifetime Partners		
	Raw	Log	(<i>n</i>)	Raw	Log	(<i>n</i>)	Raw	Log	(<i>n</i>)
Females	1.60	0.51	(187)	2.73	0.83	(188)	4.79	1.20	(188)
Males	2.92	0.80	(106)	5.17	1.12	(104)	8.35	1.62	(104)
<i>F</i> value		9.29 (<0.01)			7.69 (<0.01)			9.18 (<0.01)	
ACASI	2.26	0.67	(86)	4.52	1.01	(87)	7.05	1.45	(88)
CAPI	2.14	0.59	(103)	3.44	0.95	(103)	5.51	1.33	(102)
CASI	1.87	0.60	(104)	2.99	0.85	(102)	5.75	1.28	(102)
<i>F</i> value		0.93 (ns)			2.19 (ns)			3.03 (<0.05)	
Open	1.65	0.50	(102)	3.12	0.85	(102)	7.20	1.45	103
Closed-low	1.43	0.43	(105)	2.62	0.78	(103)	4.73	1.23	103
Closed-high	3.38	0.98	(86)	5.33	1.21	(87)	6.28	1.37	86
<i>F</i> value		14.05 (<0.001)			7.03 (<0.01)			2.35 (<0.10)	
Permissive context	1.88	0.57	(148)	2.95	0.85	(145)	5.77	1.34	(146)
Restrictive context	2.28	0.66	(145)	4.24	1.01	(147)	6.34	1.36	(146)
<i>F</i> value		2.92 (<0.10)			4.03 (<0.05)			0.52 (ns)	

Source: Tourangeau and Smith (1996).

Note: *F* values based on log transformed data.

five-year data. Finally the restrictive context elicited higher numbers of sex partners than the permissive context for all three time periods; however, this “backfire” effect was significant only for the five-year data, and the effect of context was qualified by higher-order interactions.

The data in Table 22.5 show the usual discrepancy between men and women in the reported number of sex partners. We expected this discrepancy to be reduced when the questions were self-administered. Moreover we expected the effect of self-administration to differ by sex—with males reporting fewer partners and women more partners under the two self-administered modes than under the interviewer-administered mode. Consistent with this hypothesis, the disparity between the number of sex partners reported by men and women was largest under CAPI for all three time periods; in addition CAPI yielded the highest level of reporting for the men and the lowest level for the women.

Figure 22.2 displays the means of the transformed sex partner reports for the past year by sex and mode of data collection. The pattern is similar for five-year and lifetime reports as well. However, none of the mode by sex interactions were significant; the pattern apparent in Figure 22.2 was qualified by higher-order effects involving the context variable. The convergence in the reports of men and women under the self-administered modes of data collec-

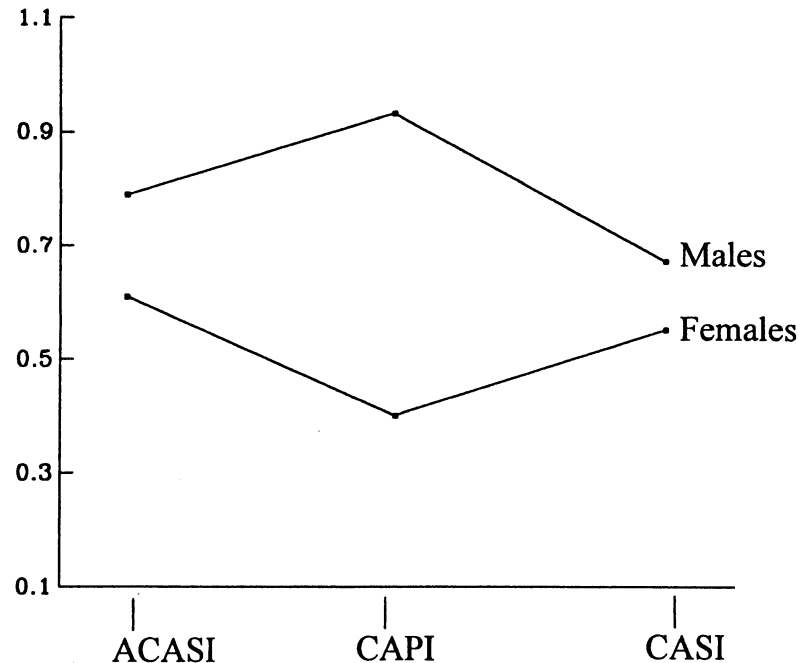


Figure 22.2. Mean reported sexual partners (log transformed) in past year, by mode and sex.

tion—CASI and ACASI—was far more pronounced among respondents who received the restrictive context items than among those who received the permissive items. Within the restrictive context groups, the mode of data collection had a large effect on the reported number of sex partners and the gap between men and women was widest under CAPI. Within the permissive context groups, the mode of data collection had relatively little effect on the number of partners reported and the disparity between the reports by men and women persisted under all three modes of data collection.

Other Sexual Behavior

In addition to the sex partners items, respondents were asked about some specific sexual behaviors, including how frequently they had had oral and anal sex over the past five years. We analyzed the proportion of respondents who reported that they had oral and anal sex at least some of the time as a function of data collection mode. A significantly larger percentage of respondents indicated they had oral sex under ACASI (75.3%) than under CAPI (51.5%) or CASI (64.6%) data collection. Similarly, a significantly higher percentage of respondents indicated they sometimes had anal sex under ACASI (20.2%) than under the other modes (4.8% and 9.8% under CAPI and CASI).

Reporting of Illicit Drug Use

The mode of data collection also had some effect on the proportion of respondents who admitted using illicit drugs. We examined reported marijuana

or cocaine lifetime use, within the past year, and during the past month. For both drugs and all three reporting periods, ACASI yielded the highest percentage of reported drug use. In general, CAPI yielded the lowest level of reporting with CASI between the other two modes. The mode difference was significant for lifetime marijuana use and marginally significant for one-year marijuana use and lifetime cocaine use. The difference between levels of reporting under ACASI and CASI was not significant for any of the drug use variables.

22.2.3 Bogus Pipeline Study

We sought to test the hypothesis that reports about sexual behavior and illicit drug use increased so strikingly with self-administration because of more or less deliberate misreporting of information to interviewers. In our final study we attempted to increase respondents' motivation to provide accurate information through a very powerful manipulation—the “bogus pipeline” procedure (Jones and Sigall, 1971). In the bogus pipeline procedure, respondents are led to believe that inaccurate answers can be detected by a physiological recording device (e.g., a polygraph) or a test performed on a biological specimen (e.g., a saliva sample). The crucial feature of the method is not that the device or test can actually detect invalid answers but that the respondents believe that it can. Reviews of the literature indicate that the technique can produce more accurate self-reported data than those obtained under standard conditions. For example, in a review of 11 studies on the effect of the bogus pipeline procedure on adolescents' self-reports about smoking, Murray et al. (1987) find an overall effect for the method.

The respondents in our third study were adult volunteers from Chicago, paid \$20 for taking part. A total of 120 were interviewed, 62 under the bogus pipeline procedure and the other 58 under control conditions. The respondents completed a preliminary “screening questionnaire,” covering such basic items as sex, age, and race. After respondents had completed this questionnaire, they were led to a second room, where the main interviewer introduced herself. The interviewer then administered the main questionnaire. The main interview covered a range of health-related topics—the respondent's smoking, drinking, and exercise habits, as well as illicit drug use and sexual behavior. Three of the items asked how many opposite-gender sex partners the respondent had in the usual three periods—the past year, past five years, and over his or her lifetime.

In the control treatment, the interviewer and the respondent sat at a desk across from each other, and the interviewer, after a brief introduction, conducted a conventional face-to-face interview. In the experimental group, the procedures were more complicated. The interviewer seated the respondent in a chair, facing a battery of machinery that included a desktop computer, a polygraph, and a microphone. The interviewer then attached four electrodes to the respondent and adjusted the microphone so that a light displayed in

response to changes in the volume of sound. The interviewer then demonstrated the equipment's apparent ability to detect lies, telling the respondent that it was necessary to "calibrate the machine" and the respondent should lie in answering some test questions. These were drawn from the screener. Without the respondents' knowledge, the experimenter in the reception area had forwarded to the interviewer their answers to the preliminary questions electronically; these were displayed on the interviewer's computer monitor. The interviewer was thus able to detect any discrepancies. After this demonstration, the main interview was conducted.

The results of the study were simple and quite dramatic—overall, 7 of the 19 sensitive items we examined showed significant differences between the bogus pipeline and control respondents, and three others showed marginally significant differences. Altogether on 15 of the 19 items the differences between groups are in the expected direction. Clearly, when they are under enough pressure, respondents are willing to admit to a number of behaviors they would otherwise deny. (For a more detailed account, see Tourangeau, Smith, and Rasinski, 1997.)

22.3 CONCLUSIONS

The most consistent finding from the studies we have summarized is the overwhelming effect of self-administration on levels of reporting sensitive behaviors. Table 22.6 displays ratios comparing the levels of reporting in interviewer-administered and self-administered conditions from the Women's Health and ACASI experiments. These ratios are analogous to the ones presented in Table 22.1. Almost without exception, respondents report fewer sex partners, less sexual activity, and less drug use when they must report their answers to an interviewer. The major exception—reduced reporting of sex partners by men—is consistent with the hypothesis that self-administration reduces self-presentation concerns. Our results are also consistent with the results of a series of studies by Turner et al. (1996a; Chapter 23). Respondents appear to withhold embarrassing information on certain topics when interviewers administer the questions. Self-administration may not eliminate misreporting due to social desirability, but it may be the single most effective means for minimizing this source of error.

The magnitude of the effect of self-administration on levels of reporting may depend on several variables, including the sensitivity of the behaviors, the level of trust of the respondents, the privacy of the interview setting, the context and format of the questions, and the respondent's race. We have already noted the effect of the degree of sensitivity. When the behavior in question is very sensitive, self-administration can have dramatic effects, doubling or tripling the proportion of respondents willing to report the behavior; when the behavior is not especially sensitive, the effect of self-administration seems negligible. Differences in sensitivity seem to account for the variations in reporting ratios

Table 22.6. Ratios Comparing Levels of Reporting by Mode of Data Collection

Contrast and Ratios		
<i>Women's Health Study</i>	<i>PAPI vs. SAQ</i>	<i>PAPI vs. CASI</i>
<i>Average number of sex partners (females)</i>		
Past year	0.97	0.72
Past five years	0.85	0.64
Lifetime	0.67	1.03
% Reporting STD	0.77	0.77
Average % of time using condom	0.84	0.70
<i>ACASI experiment</i>	<i>CAPI vs. ACASI</i>	<i>CAPI vs. CASI</i>
% Reporting anal sex	0.24	0.49
% Reporting oral sex	0.68	0.80
% Reporting marijuana use		
Past month	0.60	0.84
Past year	0.62	1.01
Lifetime	0.68	0.78
<i>% Reporting cocaine use</i>		
Past month	0.57	1.05
Past year	0.35	0.73
Lifetime	0.55	0.99
<i>Average number of sex partners (males)</i>		
Past year	1.33	1.85
Past five years	1.00	1.43
Lifetime	0.99	1.11
<i>Average number of sex partners (females)</i>		
Past year	0.61	0.68
Past five years	0.53	0.86
Lifetime	0.60	0.81

Sources: Jobe et al. (1997); Tourangeau et al. (1997); Tourangeau and Smith (1996).

Note: Sex partner data are untransformed counts.

across behaviors that are apparent in Tables 22.1 and 22.6. For a given respondent the level of sensitivity of a behavior may reflect its perceived prevalence, the degree of social approval (or disapproval), and whether the behavior is illegal or not. Sensitivity can also vary across subgroups of the population; for example, smoking is probably more sensitive for adolescents than for adults.

If the effects of self-administration reflect fears about disclosure or about the reactions of the interviewer, then they may be heightened among respondents with chronically low levels of trust in people. This hypothesis receives some

support from a study by Aquilino (1994), in which responses to an item on trust interacted with mode of data collection to affect reports of illicit drug use. Similarly one might expect self-administration to affect answers less in private settings than when other persons are present during the interview. To date, however, few studies have been able to demonstrate the effects of the privacy of the data collection setting. The Women's Health Study attempted to vary the setting of the interview experimentally and found few effects. At least two prior studies had varied the mode of data collection and recorded whether other people were present during the interview, but neither study found either main effects for the privacy measure or interactions with self-administration (Shober et al., 1992; Turner et al., 1992; see also Mosher and Duffer, 1994). In theory, the effect of the presence of other people would depend on whether the respondent has anything embarrassing to admit, whether the others present are already aware of what the respondent might report, and how the others are likely to react to the respondent's admission of something they were previously unaware of. It is one thing to admit having smoked marijuana in the presence of one's spouse, who is probably already aware of it, quite another to make the same admission in front of one's parents or children.

Our study of ACASI indicated that the effect of mode of self-administration on the reported number of sex partners depended both on the format and context of the sex partner questions. Self-administration had relatively muted effects with the open-ended versions of the sex partner questions compared with its impact with the two closed-ended versions of the items (Tourangeau and Smith, 1996). The context variable had an even clearer effect. When the restrictive items preceded the sex partner questions, the convergence between the number of sex partners reported by men and women respondents under self-administration was more marked than when the permissive items preceded them. It seems likely that the restrictive context items heightened self-presentation concerns and therefore increased the effect of self-administration on the reported number of sex partners.

The effect of self-administration may vary by demographic subgroup. For example, Aquilino and LoSciuto (1990) found that reported levels of alcohol and marijuana use were generally lower when the data were collected by telephone than in an SAQ, but as previously shown in Table 22.1, this mode difference was more pronounced among black than among white respondents. By contrast, Schober et al. (1992) found the opposite, concluding that non-minority respondents were more affected than minority respondents by self-administration in their reporting of cocaine and marijuana use. These contradictory findings about the interaction of race and mode of data collection may reflect the complex underlying relationships involved. Subgroup differences in the impact of self-administration are likely to result from differences across subgroups in the sensitivity of the behaviors in question, the perceived privacy of the different modes of data collection, or in the trust inspired by the interviewers. These differences in turn may vary over time or by region of the country.

Self-administration has its drawbacks. An interviewer is trained to recognize when respondents need help and to give such help. Self-administration can increase item nonresponse. It may also reduce the length or quality of open-ended material respondents provide. It eliminates the interviewer comments that can clarify the meaning of an answer. The great virtue of self-administration is that it permits some respondents to make admissions that would otherwise be too embarrassing.

22.3.1 Effect of Auditory Presentation

Our second experiment allows us to compare two modes of data collection—CASI and ACASI—that differ mainly in whether the items are presented only visually or both visually and aurally. The results in the lower panel of Table 22.6 suggest that ACASI may enhance the advantages of other methods of self-administration. For several items, ACASI had greater effect on reporting (relative to CAPI) than CASI did. The use of earphones may induce a greater sense of isolation and privacy than CASI does.

ACASI also helps overcome the literacy barriers to self-administration. There are several distinct sources of literacy problems. Some people never learn to read because of visual impairment or severe mental disability. Others learn to read other languages but cannot read English. By itself, ACASI does little to address such problems. Instead, ACASI—and auditory presentation more generally—is probably of most benefit with respondents who fail to learn to read because of a learning disability or lack of education.

22.3.2 Effect of Computerization

The effects of computerization on the reporting of sensitive information seem far less dramatic than those of self-administration or auditory presentation of the items. There are a few scattered effects reported in the study by Baker, Bradburn, and Johnston (1995). In the Women's Health Study, computerization reduced the rates of missing data and the variability attributable to interviewers. Our studies may have missed the full effect of computerization because our samples consisted of relatively young people, under 45 years of age. As Couper and Rowe (1996) observed, older respondents, those with little education, and those with limited prior experience with computers may be reluctant to use the new methods of data collection, which require direct interaction with the computer. Computerization may lead to increased nonresponse rates for such sample members.

Whatever its other effects, computerization has one clear virtue: It is a necessary feature for ACASI and other new technologies for collecting data. Even in the absence of clear evidence of the benefits of computerization, computers will, without doubt, continue to play a growing role in the collection of sensitive data.

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